

Steam Turbine Protection Control System

Open-Loop vs PID Control Using XSteam-Based Rankine Cycle Simulation

A Story of Steam Quality, Turbine Risk, and Why Feedback Matters

Ehab Osama Hamed
Faculty of Engineering, Benha University
May 8, 2026

One protection variable, $x_{LP,out}$, tells us whether the turbine stays in a safe dry-expansion regime or starts moving toward moisture-driven damage and costly outage risk.

The Story Starts with a Very Expensive Machine



A steam turbine is a plant-scale asset whose condition directly affects generation availability and maintenance cost.

- ▶ The turbine is huge, high-speed, complex, and expensive to maintain.
- ▶ A small control mistake can become a large thermodynamic deviation.
- ▶ That deviation can become erosion, vibration, and outage risk.

Why this matters

Failure is paid twice: repair cost first, then lost power production.

Why One Variable, x , Matters So Much



Even a small protection mistake can become a very large maintenance event because the machine itself is so large and fast.

Steam-quality meaning

$x = 1$ means dry steam

$x < 1$ means wetter steam

moisture fraction = $1 - x$

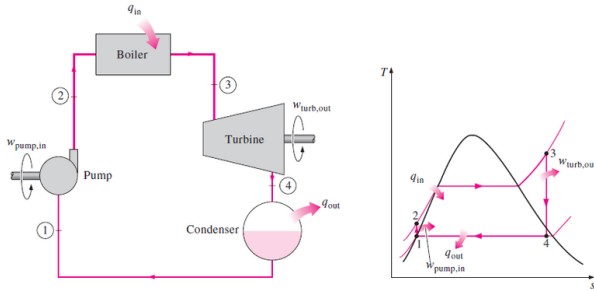
At the LP exit, a lower x means more liquid droplets in a very fast blade row.

Bad story for the turbine

Lower x means more droplet impact, more erosion risk, and less mechanical margin.

Protection viewpoint: here, x is a protection variable, not only a thermodynamic result.

The Thermodynamic Story: Where the Risk Appears



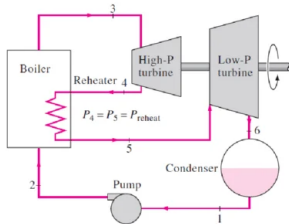
The protection problem appears at one specific place in the story: the low-pressure turbine exit.

- ▶ The boiler raises enthalpy and inlet condition.
- ▶ The turbine converts enthalpy drop into shaft work and electrical power.
- ▶ The condenser adds the back-pressure disturbance that hurts steam quality.
- ▶ The protection concern finally appears at the turbine exit.

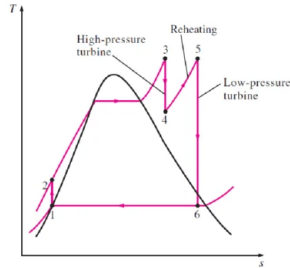
Key variable

At the LP exit, steam quality x tells us how dry or wet the steam is.

Then the Reheat Story: Why Q_{rh} Becomes the Control Handle



Reheating is a practical protection mechanism because it helps keep downstream expansion drier.



Why reheating helps

- ▶ It lifts the thermal state before low-pressure expansion.
- ▶ That increases dryness margin at the LP exit.
- ▶ So reheater duty becomes the natural manipulated input.

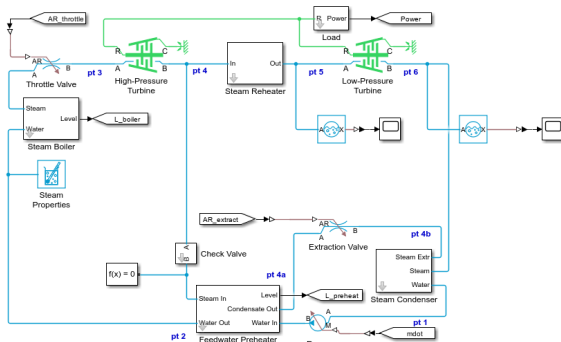
Control pair

Measured output: $x_{LP,out}$

Manipulated input: Q_{rh}

The controller moves thermal state to protect blade wetness.

The Simulation Must Look More Like a Plant Than a Textbook Sketch

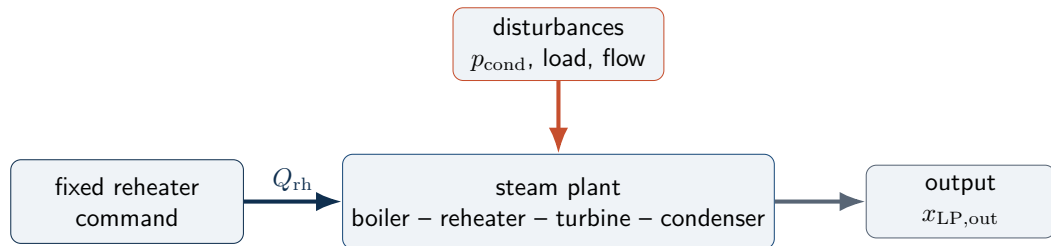


The model keeps the main physical story: boiler, turbines, reheater, condenser, valves, and the dynamics that push the LP exit toward or away from wet operation.

Why this slide matters

The plant has thermal inertia, pressure dynamics, flow changes, actuator limits, and disturbance paths. That is exactly why open-loop action is fragile and feedback becomes valuable.

Open-Loop Structure: What Is Missing?



no feedback
no correction

Open-loop looks simple because the feedback path is missing. That simplicity is exactly the weakness.

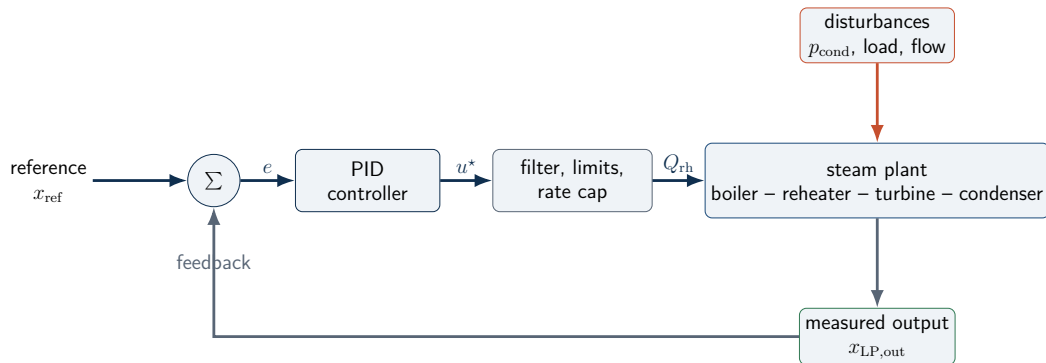
Open-loop assumption

The plant stays near nominal, so one fixed command is expected to work across the whole response.

Open-loop weakness

Disturbances move the plant, nonlinear steam behavior magnifies the mismatch, and steam quality can drift with no automatic recovery.

Closed-Loop Structure: Where Feedback Changes Everything



Feedback adds the missing path: measure, compare, and correct. That is why the controller can defend the plant against drift.

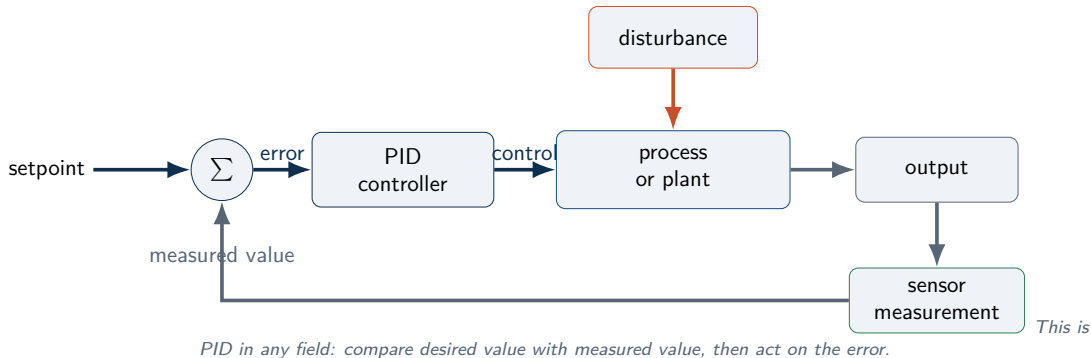
What changed

Steam quality is no longer a passive outcome. It becomes an actively regulated variable.

Why that matters

The controller reacts to error immediately instead of waiting for operator action or a protection trip.

PID in General: The Standard Feedback Idea



Generic language

Setpoint is what you want. The plant is what you control. The sensor tells you what really happened.

Controller job

The controller pushes the process toward the setpoint while keeping the response stable and practical.

What the P, I, and D Terms Actually Do

P term

$$u_P = K_p e$$

Reacts to the error right now.
More P gives stronger immediate action, but too much P can make the response oscillatory.

I term

$$u_I = K_i \int e dt$$

Removes steady offset by accumulating error over time.
Too much I can create windup and slow oscillation.

D term

$$u_D = K_d \frac{de}{dt}$$

Reacts to how fast the error is changing. It adds damping, but too much D makes the controller sensitive to noise.

Combined control law

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

How to Read the PID Parameters Before Any Project Details

Main gains

- ▶ K_p : aggressiveness to present error.
- ▶ K_i : speed of steady-error removal.
- ▶ K_d : damping and anticipation.

If they are too low

The loop becomes slow, weak against disturbances, and may leave a noticeable offset.

Practical parameters

- ▶ filter time constant
- ▶ saturation limits
- ▶ rate limit
- ▶ anti-windup logic

If they are too high

The loop becomes aggressive, noisy, oscillatory, and harder on the actuator.

Tuning mindset

Good tuning is always a compromise between speed, stability, noise sensitivity, and actuator protection.

How the MATLAB and XSteam Simulation Was Built

Step 1: define the plant

- ▶ Nominal pressure, temperature, condenser pressure, and flow.
- ▶ Dynamic time constants for boiler, reheater, condenser, and flow.
- ▶ Protection thresholds $x_{\min}$ and x_{crit} .

Step 2: excite the plant

- ▶ Boiler, load, condenser, and flow disturbances are applied as long smooth pulses.
- ▶ The condenser disturbance is intentionally quality-critical.
- ▶ Open-loop and PID are simulated on the same plant.

Step 3: judge performance

- ▶ Track x versus the nominal reference.
- ▶ Compute settling time and IAE.
- ▶ Measure unsafe and critical exposure time.
- ▶ Compare final quality and net power recovery.

Simulation philosophy

The model is simple enough to explain, but rich enough to show the real lesson: disturbances move the plant, and feedback must reshape the response to protect the turbine.

Only the Control Lines That Carry the Main Idea

```
1 x_f = x_f + dt * (x_true - x_f) / max(ctrl.tau_x, dt);
2 dx_f = (x_f - x_f_prev) / dt;
3 e = p.x_ref - x_f;
4 if ~((res.Q(k) >= p.Q_hi - 1e-6 && e > 0) || ...
5     (res.Q(k) <= p.Q_lo + 1e-6 && e < 0))
6     x_int = clamp(x_int + e * dt, -ctrl.int_lim, ctrl.int_lim);
7 end
8 Q_target = p.Q_nom + ctrl.Kp * e + ctrl.Ki * x_int - ctrl.Kd *
9     dx_f;
10 Q_target = clamp(Q_target, p.Q_lo, p.Q_hi);
11 Q_smooth = res.Q(k) + dt * (Q_target - res.Q(k)) / ctrl.tau_q;
12 Q_next = clamp(rate_limit(res.Q(k), Q_smooth, ctrl.dQ_max), p.
13     Q_lo, p.Q_hi);
```

These are the key lines to explain: measure quality, compute error, build PID action, then protect the actuator with limits and rate control.

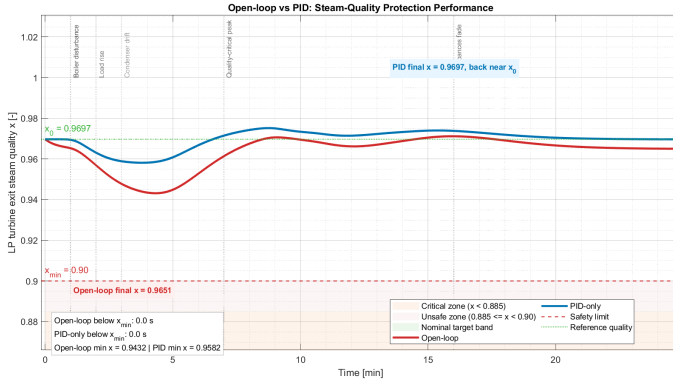
How to explain the code

- ▶ Filter the measured quality before differentiation.
- ▶ Compute the error between desired and measured x .
- ▶ Use PID to ask for a stronger or weaker reheater command.
- ▶ Clamp and rate-limit the command to keep it physically realistic.

Actual tuning from the script

$K_p = 6.5$, $K_i = 0.012$, $K_d = 0.050$,
 $\tau_x = 12$ s, $\tau_q = 10$ s, and
 $dQ_{\max} = 0.020$.

The Main Evidence: Steam Quality Response

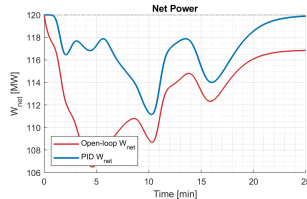
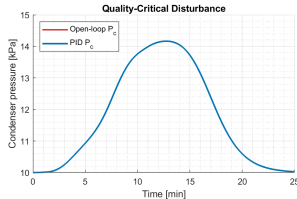
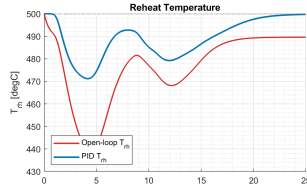
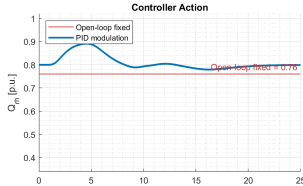


This is the central comparison: PID keeps more margin and recovers closer to the nominal quality.

- ▶ The open-loop trace falls farther from the reference quality.
- ▶ The PID trace reacts sooner and stays closer to the target band.
- ▶ That means less moisture risk under the same disturbance.

What to say aloud: PID does not remove the disturbance, but it stops the disturbance from owning the quality trajectory.

The Supporting Evidence: PID Changes the Plant Reaction



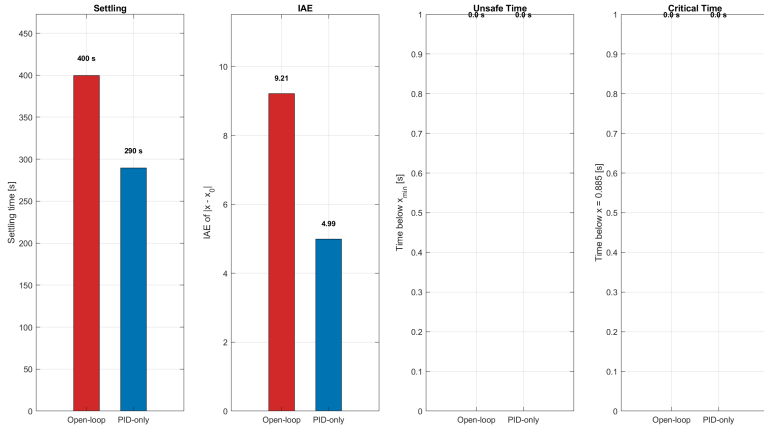
quality because it reshapes both the thermal input and the plant response.

PID improves

- ▶ Q_{rh} is modulated, not fixed.
- ▶ Reheat temperature recovers with better damping.
- ▶ The condenser-pressure disturbance is rejected more effectively.
- ▶ Net power recovers more strongly under PID.

Control power: good control is visible in how the input changes the plant trajectory.

The Dashboard Closes the Argument in One Glance



The dashboard compresses the comparison into decision-ready metrics: faster settling and lower integrated error under PID.

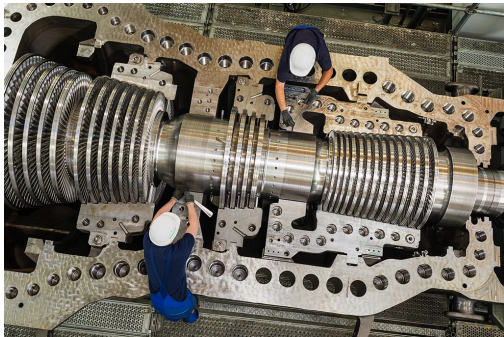
What improved

- ▶ Settling time: 400 s → 290 s
- ▶ IAE: 9.21 → 4.99
- ▶ Unsafe time: 0.0 s in both cases
- ▶ Critical time: 0.0 s in both cases

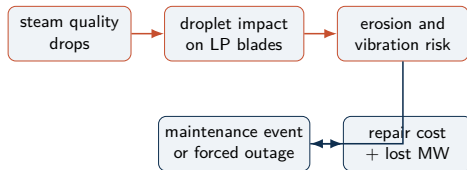
Important insight

Even when both cases stay above the threshold, PID still matters because it preserves more margin and reaches recovery faster.

If Control Fails, the Plant Pays Twice



A turbine problem is rarely isolated. Mechanical damage creates maintenance cost while lost generation creates operational cost.



Plant-level message

Good control protects availability, maintenance budget, and operating confidence.

Final Message: This Is Why Control Matters

- ▶ Steam quality x is a protection variable, because a thermodynamic deviation at the LP exit can become a mechanical damage mechanism.
- ▶ The steam turbine is large, fast, complex, and expensive, so small errors can create disproportionately large consequences.
- ▶ Open-loop operation can look acceptable at nominal conditions, but it loses authority when real disturbances move the plant.
- ▶ PID feedback turns $x_{LP,out}$ into an actively regulated variable and gives the plant a stronger protection margin.



Takeaway

Control bridges thermodynamics and asset protection: it keeps the cycle productive while keeping the turbine safer.

Next step

Share the final MATLAB file later and this deck can use your exact controller lines everywhere.

Your Opinion Matters to Me



Feedback form

Scan this QR code to share your opinion, suggestions, or ideas for improving the work.



GitHub folder

Scan this QR code to open the project files, figures, and supporting material on GitHub.

A personal note

I sincerely appreciate your feedback. Your comments, questions, and criticism will help me improve both the model and the presentation.

Thank You

Your time, attention, and feedback mean a lot to me.

Questions and discussion are warmly welcome.



كلية الهندسة - بنها
FACULTY OF ENGINEERING - BENHA

With appreciation for your opinion, suggestions, and support.